NumPy Assignment 1.3

1. Explain the difference between arr[:,0] and arr[:,[0]] with an example.

In NumPy, both arr[:,0] and arr[:,[0]] are used to select the first column of a two-dimensional array, but they return different types of outputs. The expression arr[:,0] selects all rows and the first column, returning a one-dimensional (1D) array.

```
import numpy as np

arr = np.array([[10, 20, 30],

            [40, 50, 60],

            [70, 80, 90]])

print(arr[:,0])

> array([10, 40, 70])
```

On the other hand, arr[:,[0]] also selects all rows and the first column, but because the column index is provided as a list ([0]), NumPy returns a two-dimensional (2D) array.

```
import numpy as np

arr = np.array([[10, 20, 30],

            [40, 50, 60],

            [70, 80, 90]])


arr[:,[0]]

> array([[10],

  [40],

  [70]])
```

2. Explain the difference between Vectorization and Broadcasting in Numpy with proper example.

Vectorization is the process of performing operations on entire NumPy arrays without writing explicit loops in Python. Instead of processing elements one by one using a for loop, NumPy applies the operation to all elements simultaneously using highly optimized C implementations.

Example:

```
arr1 = np.array([1, 2, 3, 4, 5])
print(arr1*2)
> [ 2  4  6  8 10]
```

Broadcasting, on the other hand, is a mechanism that allows NumPy to perform arithmetic operations on arrays of different shapes. NumPy automatically expands the smaller array so that its dimensions become compatible with the larger array.

Example:

```
A = np.array([[1,2,3],[4,5,6]])
B = np.array([10,20,30])
print(A + B)
>  [[11 22 33]
 [14 25 36]]
```